

APPENDIX C
FOR
Mobile Storage System
AT
PITTSBURGH ANGB, PENNSYLVANIA
BUILDING 110
CSN NPT501

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1. SCOPE

1.1. GENERAL SCOPE

This appendix establishes the minimum performance, design, fabrication, installation, and test requirements for a mobile storage system at Pittsburgh ANGB, Building 110. The work shall be completed in accordance with this appendix, the cover PD, and Air Force (AF) Drawings NPT501-01. This appendix is intended to cover all equipment, materials, and labor necessary to complete installation of the system.

2. APPLICABLE DOCUMENTS

Applicable documents shall be in accordance with Section 2 of the cover PD.

3. REQUIREMENTS

3.1. SYSTEM DESCRIPTION

The mobile storage system shall conform to the requirements of NPT501-01.

3.2. CHARACTERISTICS

The mobile storage system shall conform to the following requirements.

3.2.1. SYSTEM OPERATION

Each carriage shall have a crank mechanism as shown on AF drawing NPT501-01.

3.2.2. CONFIGURATION REQUIREMENTS

The mobile storage system shall be located and configured as shown on AF Drawing NPT501-01.

3.2.3. STRUCTURE AND INSTALLATION

- Carriages
 - Capable of supporting a minimum load of 1,000 pounds per carriage foot.
 - Not deflect greater than .5 percent of distance between supports.
 - Not exhibit permanent deformation.
- Frame
 - Be of aluminum or steel channel, welded or bolted together to support a fully loaded shelving section.
 - Have cross bracing.
- Bumpers

- Be mounted on frame to ensure spaced appropriately to minimize carriage contact.
- Stationary platforms
 - Be of the same construction and height as the carriages.
 - Anchored to rails or floor.
 - Require no drive train.
- Shelving sections
 - Pallet rack type C and D inserts shall conform to appendix B and interface with the mobile storage system in building 110.
- Face panels
 - Cover entire height and width of each shelving section.
 - Be 1/8 inches thick.
 - Fastened to both ends of shelving sections.
 - Provide rigid mounting location for drive handle, locks and card holders.
- Row identification
 - Be installed on face panels at eye level.
 - Be at least five inches wide by three inches high.
 - Provide blank cards for personnel assignment.
- Rail
 - Be installed on load bearing grout which spans entire width and length of rail.
 - Be of steel material.
- End stops
 - Be installed to prevent carriage run off.
 - Be securely anchored to rail or floor.
- Rail skirts
 - Be installed to cover unfinished edges of floor, rail and ramp.
- Elevated flooring
 - Resin deck or equivalent.
 - Minimum of 1/2 inch thick.
 - Exterior sealant used Acryseal or equivalent.
 - Capable of supporting 75 pounds per square foot.
 - Installed between and flush with top of rails
 - Adjacent sheets of subflooring shall be shimmed along the full length of the common side.
- Ramps
 - Be installed on all sides used to access mobile aisle.

- Be of steel material.
- Be capable of supporting 75 pounds per square foot.
- Not extend past the end of the carriages into the aisle.
- Grade not to exceed 10 percent.
- Drive chain
 - Connect a set of gear reduction sprockets from the crank shaft to the drive shaft.
- Bearings
 - Be permanently lubricated, securely mounted, and properly housed for protection from physical damage.
- Wheels
 - Not lose contact with rails.
- Guidance system
 - Prevents dislodgement or derailment from rails.
- Safety devices
 - Have anti-movement lock to prevent carriage displacement.
 - Have anti-tip device to prevent tipping during movement.
 - Have system lock devices.

4. QUALITY ASSURANCE PROVISIONS

4.1. INSPECTIONS

The system shall be inspected to verify that it meets all the requirements of sections 1, 2, and 3 of this appendix and the cover purchase description. These inspections shall be completed as part of the systems checkout test and again as a part of the quality conformance test.

4.2. TESTS

The following tests shall be performed on the system as part of the system checkout test and again as a part of the quality conformance test.

No.	Description	Result
1.	A functional test of 24 hours in three eight hour increments shall be made on the system. It shall be tested randomly over this period. Test each crank by turning it both clockwise and counterclockwise. If the carriage does not move in the desired direction, the system fails the test.	☐ Pass ☐ Fail

2. Maximum loads shall be placed randomly in the system. After not less than four hours, the test loads shall be removed and the system examined for signs of deformation. If any component of the system exhibits plastic or elastic deformation outside of the approved parameters, the system fails the test.

☐ Pass
☐ Fail
3. Test each anti-movement lock. If a carriage can be moved with a lock activated, the system fails the test.

☐ Pass
☐ Fail